

ALEX KERSELL

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PROFESSIONAL SUMMARY

Electrical engineer who likes building things fast and getting them working end-to-end — from a handheld ultrasonic levitation device built solo from first principles to Python tools that turned tedious manual work into a few lines of code. Comfortable moving across hardware, firmware, and software, and drawn to startup environments where scrappy prototyping, tight iteration loops, and direct ownership matter more than a fixed job description.

SKILLS

Rapid Prototyping | Python | C++ | Java | Altium (PCB Design)
Embedded Systems | Digital Signal Processing | Human-Centered Design | Hardware/Software Integration | Self-Directed Problem Solving

BUILDER HIGHLIGHTS

- Senior Capstone Project** | *Handheld Acoustic Levitation Device* 09/2025 – 05/2026
- Took a handheld ultrasonic transducer array from concept to working prototype solo, levitating and manipulating 5mm polystyrene spheres
 - Derived transducer placement and phasing from standing wave theory rather than off-the-shelf references, iterating quickly through several hardware revisions
 - Wrote the full control stack in Python to drive real-time levitation and movement, closing the loop between hardware and software myself
- Intel Curated Project** | *Making Magic of the Mundane* 01/2023 – 05/2023
- Went from open-ended prompt to working prototype in one semester: conceptualized, designed, and built an accessible note-taking alternative
 - Made fast design trade-offs under time constraints typical of a short sprint-style build
- Sponsored Education Practicum** | *Community Vision Project* 09/2023 – 12/2023
- Ran user testing and heuristic analysis directly with end users, then translated findings into concrete product recommendations

WORK HISTORY

- PV/BESS Engineering Intern** | *RRC Internship* 06/2025 – 12/2025
- Built a Python tool from scratch to automate geographic bore length estimation, replacing a manual process
 - Performed ampacity analysis and produced markups for 10% IFC drawings, moving fast across multiple concurrent site designs
- Low-Voltage Power Distribution Intern** | *Eaton Internship* 06/2024 – 12/2024
- Designed, prototyped, and tested low-voltage power distribution units (PDUs) end-to-end
 - Worked across cross-functional teams to validate products and turn around documentation quickly
- Specialty Department Team Member** | *Whole Foods* 05/2023 – 08/2023
- Handled fast-paced, hands-on operational work across multiple departments

EDUCATION

Bachelor of Science in Electrical Engineering — Minor in Innovation 05/2026
University of Portland — Portland, OR